

HARDIK DARJI

SENIOR EMBEDDED SOFTWARE ENGINEER

Embedded C | CAN / J1939 | HYDAC TTC | CANTATA | Machine Validation

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PROFESSIONAL SUMMARY

Senior Embedded Software Engineer with 4+ years of experience in off-highway construction machinery, specializing in Embedded C, CAN/J1939 communication, HYDAC TTC controller platforms, software debugging, unit testing, machine validation, and functional testing. Hands-on experience developing and modifying C software for Paver and Compactor machines, debugging CAN communication using PCAN-Explorer, monitoring I/O with HYDAC MST tools, and leading CANTATA unit-testing activities across 40+ modules in four projects. Also supports third-party telematics coordination, issue escalation, and internal user guidance.

CORE EXPERTISE

Embedded Software Development • CAN/J1939 • HYDAC TTC Controllers • CANTATA Unit Testing • Machine Validation • Functional Testing • PCAN-Explorer • HYDAC MST • Real-time Debugging

PROFESSIONAL EXPERIENCE

Senior Engineer | Ammann India Pvt. Ltd. | Ahmedabad, India | 02/2025 – Present

- Develop and modify Embedded C software for Paver and Compactor construction machinery using HYDAC TTC controller platforms.
- Work with CAN and J1939 communication for machine software development, debugging, and validation.
- Debug machine software and CAN communication using PCAN-Explorer traces and HYDAC MST tools for I/O monitoring and troubleshooting.
- Own CANTATA unit-testing activities across four projects, covering two Paver and two Compactor projects and more than 40 software modules.
- Achieved 100% coverage of the defined safety-related software scope requiring testing across the assigned projects.
- Perform machine validation and functional testing to verify software behavior on target machines.
- Coordinate with the third-party telematics implementation team, communicate technical issues, support issue resolution, and guide internal personnel on telematics system usage.

Engineer | Ammann India Pvt. Ltd. | Ahmedabad, India | 02/2024 – 02/2025

- Worked on Embedded C software and CAN/J1939-based applications for Paver and Compactor/Roller machines.
- Supported software development, debugging, testing, and machine validation activities for construction machinery.
- Worked with HYDAC TTC controller environments and machine communication interfaces.

Junior Engineer | Ammann India Pvt. Ltd. | Ahmedabad, India | 06/2023 – 02/2024

- Supported Embedded C software development and CAN communication for industrial automation and construction machinery.
- Assisted with software debugging, testing, and machine-level validation activities.

Graduate Trainee Engineer | Ammann India Pvt. Ltd. | Ahmedabad, India | 06/2022 – 06/2023

- Gained hands-on experience in embedded software development, communication protocols, development tools, and software testing.
- Worked with Git for version control and CANTATA for unit testing and software reliability activities.
- Built a strong foundation in Embedded Systems and industrial automation.

Internship Trainer | Shriji TechnoAspire Pvt. Ltd. | Gandhinagar, India | 01/2022 – 04/2022

- Worked with Arduino, ESP8266, and ESP32 development boards and integrated ultrasonic, IR, and PIR sensors.
- Implemented UART, SPI, and I2C communication for embedded projects and sensor data exchange.

TECHNICAL SKILLS

Programming: Embedded C, Python

Communication Protocols: CAN, J1939, SPI, UART, I2C

Controllers & Platforms: HYDAC TTC controller platforms, Arduino, ESP8266, ESP32

Testing & Validation: CANTATA Unit Testing, Machine Validation, Functional Testing

Debugging & Development Tools: PCAN-Explorer, HYDAC MST, Eclipse IDE, Git, VS Code

Core Competencies: Embedded Systems, CAN/J1939 Debugging, Software Development, Protocol Implementation, Real-time Debugging

EDUCATION

Bachelor of Engineering in Electronics and Communications Engineering | Government Engineering College, Gandhinagar, India | 06/2022 | CGPA: 7.55

Diploma in Electronics and Communications Engineering | Government Polytechnic, Ahmedabad, India | 07/2019 | CGPA: 7.12

LANGUAGES

Gujarati (First Language) | Hindi (Proficient) | English (Elementary)